

Developmental Dyslexia and ADHD in Children: Cognitive and Neuroanatomical Mechanisms of Co-occurrence

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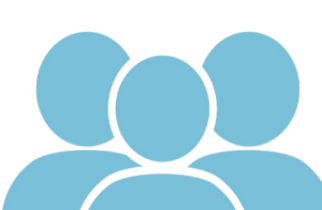
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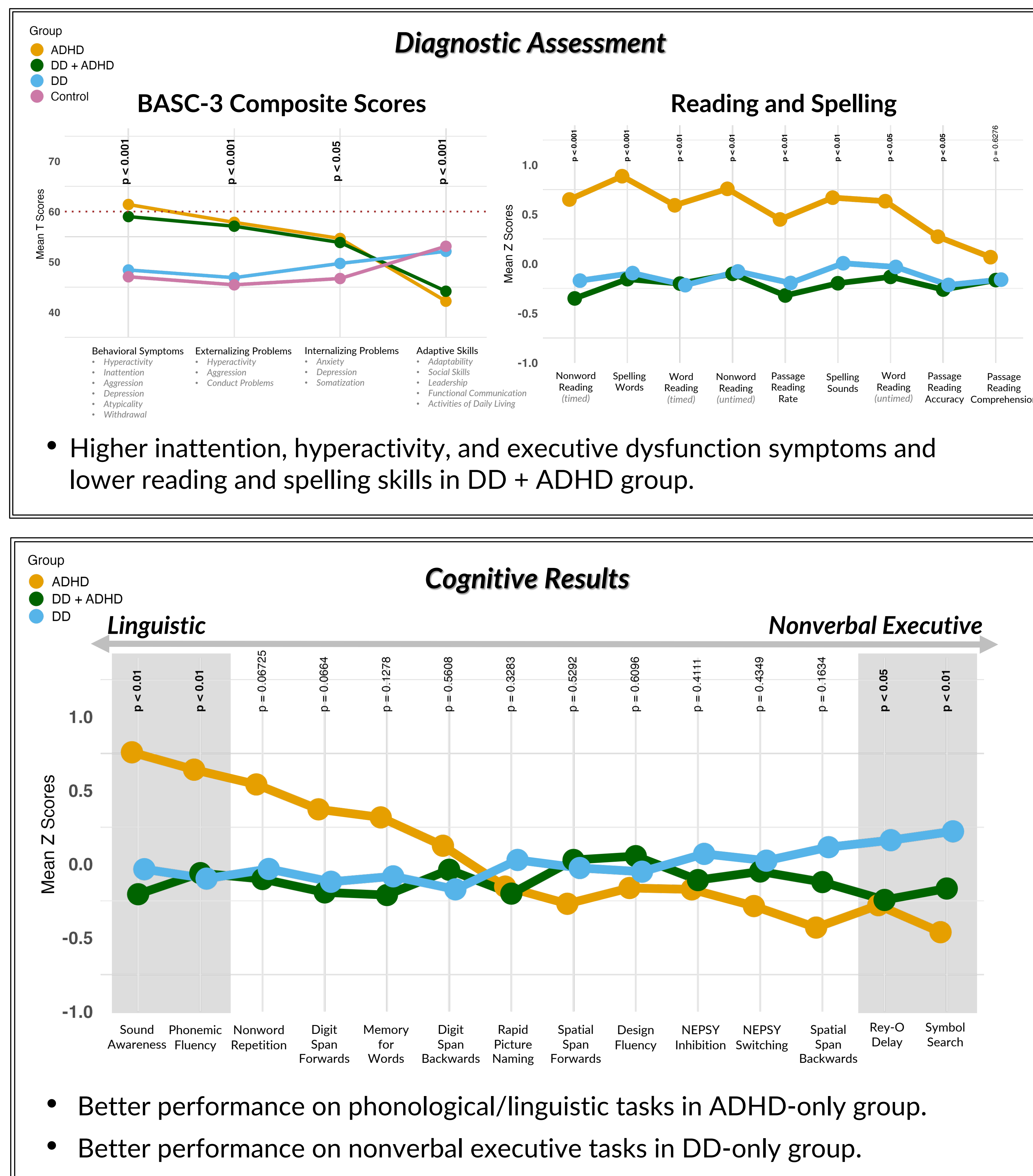
Background & Research Questions

- Developmental Dyslexia (DD) and Attention-Deficit/Hyperactivity Disorder (ADHD) are common neurodevelopmental conditions that frequently co-occur (25–40%). The neurocognitive mechanisms of this co-occurrence are not well understood.
- ADHD:** Difficulties in attention and cognitive/emotional control. Brain differences in attentional and executive networks, including cingulo-fronto-parietal and fronto-striatal circuits.
- DD:** Difficulties in decoding written words. Brain differences in left-lateralized language and phonology-related networks, including temporo-parietal, occipitotemporal, and inferior frontal regions.
- Q1 (cognition):** Do children with DD + ADHD show shared cognitive differences (phonological and decoding difficulties seen in DD alongside executive difficulties seen in ADHD)?
- Q2 (neuroanatomy):** Is DD + ADHD associated with connectivity disruptions in language and phonology-related white matter pathways as well as in executive/attention-related pathways?

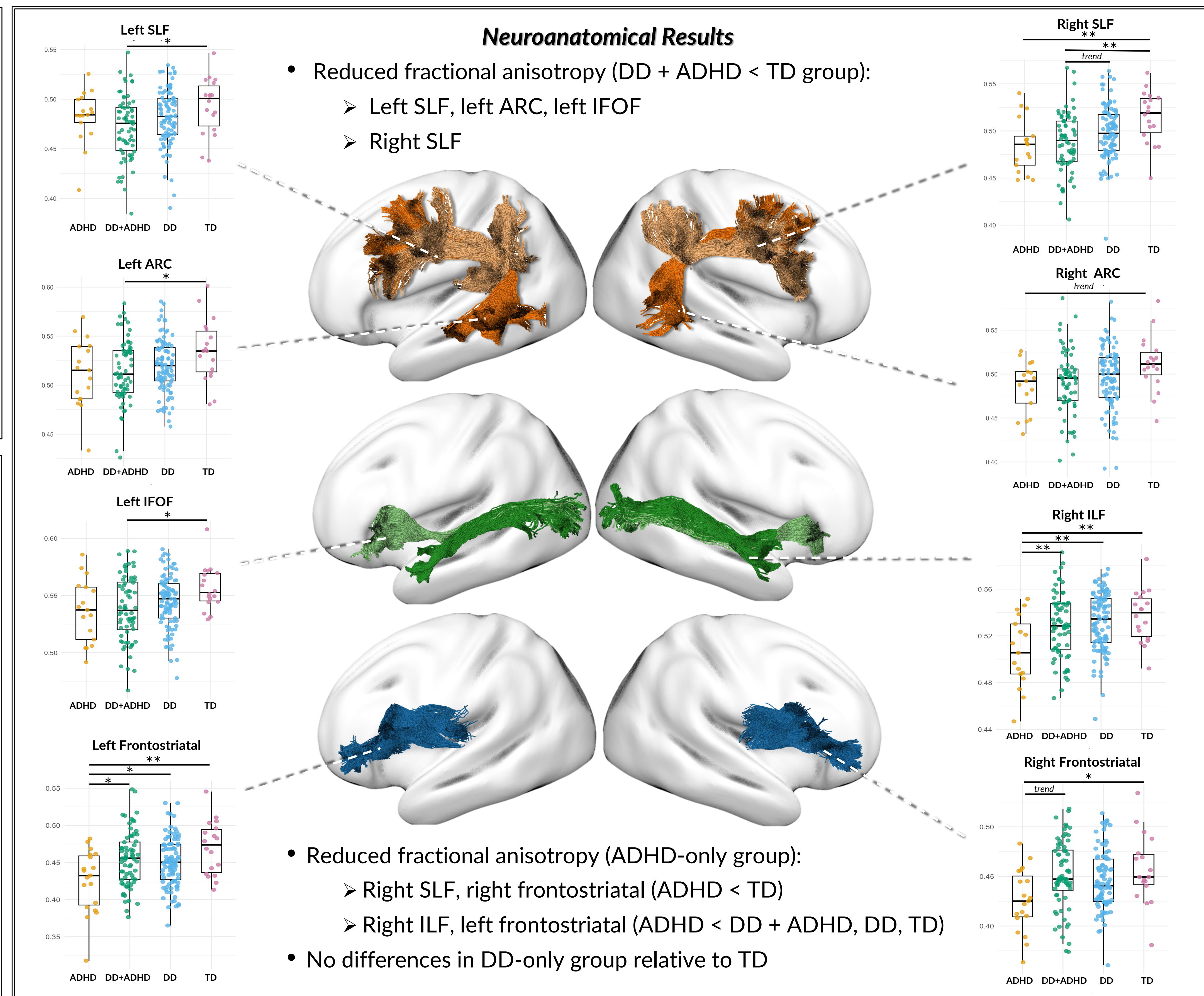
Participants & Group Definitions

- MRI + multidisciplinary assessment in **207** native English speaking, school-age children between 7–16 years (mean age ~ 11.5 years).
- Diagnostic decisions: parent/teacher and clinician reports as well as comprehensive assessment and multidisciplinary consensus conference at UCSF Dyslexia Centre.

	DD		ADHD	
	Parent/teacher reports of reading difficulties, and reading scores <25 %ile and discrepant with other abilities	Typical IQ, and no other explanations (bilingual, limited English instruction)	Parent/teacher reports of inattention/hyperactivity and BASC-3 matching criteria	Behavioral observations during testing and no better explanations (ASD, anxiety)
 99 DD 50 Male 91 R-handed	✓	✓	✗	✗
 20 ADHD 15 Male 18 R-handed	✗	✗	✓	✓
 68 DD + ADHD 43 Male 58 R-handed	✓	✓	✓	✓
 20 TD 7 Male 19 R-handed	✗	✗	✗	✗



Results



Materials & Methods

Neuropsychological Measures

- A comprehensive test battery assessing reading, spelling, phonology, short term and working memory, executive functioning, processing speed
- Behavior Assessment System for Children-Third Edition (BASC-3) parent questionnaire

Statistical Analyses

- Behavioural:** ANCOVAs to compare groups, controlling for age, sex, IQ
- Neuroimaging:** ANCOVAs to compare groups; outcome variables: mean fractional anisotropy; covariates of no interest: Sex, age, IQ, eTIV, mean frame displacement

MRI Acquisition & Image Processing

- 3T Siemens Prisma (T1w & multi-shell diffusion; b=0/500/1000/2500, 0/30/64/96 directions)
- Whole-brain tractography with CSD in MRtrix3 in qsiPrép

DD	ADHD
A priori tracts (selected using pyAFQ)	
ASSOCIATION FIBERS	
AF, posterior AF, SLF	SLF, Frontal Aslant Tracts
COMMISSURAL FIBERS	
Posterior Corpus Callosum	Anterior Corpus Callosum
PROJECTION FIBERS	
Frontostriatal tracts, ATR	Frontostriatal tracts, ATR

Conclusions

- Overlap between DD and DD + ADHD groups occurs on verbal tasks involving executive functions, such as processing speed (rapid naming) and digit backwards (working memory), with the ADHD + DD showing more severe change in left-lateralized, language-related white matter tracts.
- The ADHD-only group showed greater difficulties in nonverbal executive tasks and greater changes in bilateral fronto-striatal and dorsal right hemisphere tracts.
- The diagnostic co-occurrence of ADHD and DD might reflect preferential involvement of specific neural networks.

Scan for References

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